

Saastha Vasan

AI Security Researcher

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Profile

AI security researcher specializing in **malware analysis**, **reverse engineering**, and **ML-driven threat detection**. My work combines top-tier security and ML research with industry experience building malware detection pipelines, XDR anomaly detection methods, and red-team assessment infrastructure. I focus on translating research into practical systems for malware investigation, cyber threat intelligence, and vulnerability analysis.

Education

- Sept 2021 – **Doctor of Philosophy (Ph.D.) in Computer Science**, *University of California, Santa Barbara*
Present Advisors: Prof. Giovanni Vigna and Prof. Christopher Kruegel. Expected Graduation: July 2026.
- July 2016 – **Bachelor of Technology in Computer Science**, *Amrita Vishwa Vidyapeetham*, Kerala, India
July 2020

Industry Experience

- July 2025 – **Ph.D. Intern**, *Cisco Systems, Inc. (Talos Group)*, Irvine, California, USA
- Sept 2025
- Developed a large-scale .NET malware detection pipeline over 5M+ malicious and benign samples, covering feature engineering, model training, and evaluation, achieving a 10% relative F1 improvement over prior academic baselines.
 - Designed cross-platform XDR anomaly detection using Markov-chain process models and ancestry trees to identify anomalous execution patterns across macOS and Windows.
- Dec 2020 – **Infosec Engineer**, *Aspirify Pvt. Ltd.*, New Delhi, India
- July 2021
- Built red-team assessment modules for RCE, lateral movement, and N-day vulnerability simulation using C, C++, C#, and Python.
 - Built modular exploit payloads and post-exploitation scripts that expanded the red-team framework's coverage for enterprise assessments.

Research Experience

- Sept 2021 – **Graduate Researcher**,  *Seclab, UC Santa Barbara*, Santa Barbara, CA
- Present
- Led research on **MalwarePT**, a binary-level foundation model for malware analysis, covering pretraining, downstream evaluation, and robustness analysis.
 - Built **ML/LLM pipelines** that map malware behavior to malicious capabilities, convert attack evidence into CTI, and support automated vulnerability root-cause analysis.
- Mar 2020 – **Research Intern**,  *Seclab, UC Santa Barbara*, Santa Barbara, CA
- Sept 2020
- Researched AI methods for **malware post-detection** analysis.
 - Designed an AI-driven framework for identifying malicious capabilities in Windows malware using static analysis and neural networks.
- Oct 2016 – **Student Researcher**, *Security Lab, Amrita Vishwa Vidyapeetham*, Kerala, India
- Mar 2020
- Analyzed malware samples and documented technical findings for reverse-engineering and threat-analysis research.
 - Captained **team bi0s** in Capture The Flag (CTF) competitions, helping the team become India's top-ranked CTF team while contributing to reverse-engineering challenges.

Publications

- Under Review **SoK: Challenges and Opportunities for AI in Converting Attack Evidence to Shareable Intelligence**
- Conducted a literature review and practitioner survey to define a **Cyber Threat Intelligence (CTI) generation pipeline**, and evaluated the feasibility of **LLMs** for converting attack evidence into shareable intelligence. **(1st Author)**
- Under Review **MalwarePT: A Robust Binary-Level Foundation Model for Malware Analysis**
- Proposed **MalwarePT**, a general-purpose foundation model for malware analysis that achieves superior **generalizability** and **performance** across different malware analysis tasks. **(1st Author)**
- 2024 **DeepCapa: Identifying Malicious Capabilities in Windows Malware**, *40th Annual Computer Security Applications Conference (ACSAC 2024)*
- Developed an **AI-driven static-analysis framework** that maps Windows malware API behavior to **MITRE ATT&CK** capabilities. **(1st Author)**
- 2024 **Invisible Image Watermarks Are Provably Removable Using Generative AI**, *38th Conference on Neural Information Processing Systems (NeurIPS 2024)*
- Demonstrated a family of **regeneration attacks** using generative AI to provably remove invisible watermarks, outperforming traditional removal methods. **(3rd Author)**
- 2023 **COLUMBUS: Android App Testing Through Systematic Callback Exploration**, *45th International Conference on Software Engineering (ICSE 2023)*
- Built a **callback-driven Android fuzz-testing system** that improves app code coverage through systematic callback exploration. **(2nd Author)**

Additional Experience

- 2023 – 2025 **DARPA AI Cyber Challenge (AIxCC)**,  *Team Shellphish*
- Core member of Team **Shellphish** during the two-year DARPA AIxCC effort; contributed to automated root-cause analysis and vulnerability patching systems. The team secured a top-7 semi-final finish, won **\$2 million**, and advanced to the finals.
- 2023 – Present **Co-Chair**,  *NSF-funded ACTION Institute Student Executive Council*
- Lead **internship recruitment**, coordinate guest **speaker sessions**, foster **research collaborations** across 11 universities, and drive **outreach programs** introducing AI and security to high school students.

Achievements

- **Academic Excellence Fellowship (2021)**, University of California, Santa Barbara
- **Graduated Magna Cum Laude (2020)**, Amrita Vishwa Vidyapeetham
- **Student Excellence Award (2018, 2019)**, Amrita Vishwa Vidyapeetham

Technical Skills

- Languages Python, C, C++, C#, x86 Assembly
- Research Malware analysis, reverse engineering, binary analysis, cyber threat intelligence, vulnerability root-cause analysis
- ML/Sec Malware detection, binary representation learning, transformer pretraining, LLM fine-tuning, LoRA/QLoRA, LLM evaluation
- Sys/Sec Windows/Linux/macOS, Windows internals, PE format, .NET malware, EDR/XDR telemetry, MITRE ATT&CK, static/dynamic analysis
- Tools IDA Pro, Binary Ninja, Ghidra, CodeQL, YARA, CAPA, GDB, WinDbg
- Infra/Data PyTorch, Hugging Face, Docker, Git, PostgreSQL, MongoDB